Effect of Ethanolic Seed Extract of *Dacryodes Edulis* on Kidney Function in Carbon Tetrachloride Induced Renal Toxicity in Rats

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ABSTRACT

This study determines serum creatinine, urea, electrolytes and kidney histology in carbon tetrachloride (CCl_4) induced renal toxicity in rats. *Dacryodes edulis* (*DE*) seeds were collected and ethanolic extraction performed following standard laboratory procedures. Ninety adult male rats (weighing 180-200g) were distributed into 3 groups; normal control, CCl_4 intoxication (intraperitoneal) without treatment (1ml/kg body of CCl_4; mixed in 1:1 olive oil every 72 hours for 3 doses) and the test group (CCl_4 exposed treated with 1500 mg/kg *DE* extract). Six rats were sacrificed for blood collection by days 0, 7, 14, 21 and 28 post-treatment for estimation of serum urea, creatinine, sodium, potassium, chloride and bicarbonate ions. The kidney was harvested and fixed in 10% formalin by day 28 for histological processing. Data were analyzed using Statistical Package for Social Science software using one-way analysis of variance followed by Least Significant Difference test for comparison between groups at $p < 0.05$. CCl_4 resulted in significant increase in serum urea and creatinine at baseline and DE treatment significant decrease in serum urea by day 7 and serum creatinine by 14 days post-treatment. Also, CCl_4 exposure caused a significant increase in serum Na^+, K^+ and Cl^- and HCO_3^- at baseline. Treatment with DE did not significantly mitigate these electrolytes compared to control or untreated groups. The histological slides of group B and C showed vacuolations and defective glomeruli as compared to the normal glomeruli in the control. These findings showed *DE* to have a weak action in mitigating CCl_4 induced renal damage.

Keywords: Plant medicine, *Dacryodes edulis*, Renotoxicity, Kidney function, Carbon tetrachloride,

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1. INTRODUCTION

Plants are important sources of medicines and presently about 25% of pharmaceutical prescriptions in the United States contain at least one plant-derived ingredient [1]. In the last century, roughly 121 pharmaceutical products were formulated based on the traditional knowledge obtained from various sources. In many developing countries, a large proportion of the population relies on traditional practitioners and their armamentarium of medicinal plants in order to meet health care needs. Although modern medicine may exist side-by-side with such traditional practice, herbal medicines have often maintained their popularity for historical and cultural reasons. Such products have become more widely available commercially [1].

A good number of plants have traditionally been used by Nigerians and elsewhere because they are natural products, environmentally friendly, easily available, cheap and curative than many sub-standard orthodox medicines imported into the country today. Of interest in the present study is the African pear fruit (*Dacryodes edulis; DE*). *DE* is an evergreen tree attaining a height of 18–

40 m in the forest but not exceeding 12 m in plantations. It has a relatively short trunk and a deep, dense crown. The bark is pale gray and rough with droplets of resin. The leaves are a compound with 5-8 pairs of leaflets. The upper surface of the leaf is glossy. The flowers are yellow and about 5 mm across. They are arranged in a large inflorescence. The fruit is an ellipsoidal drupe which varies in length from 4 to 12 cm. The skin of the fruit is dark blue or violet, whereas the flesh is pale to light green [2].

DE is a versatile plant in African ethno medicine, as its various parts are employed to treat several diseases. Its bark has long been used to cicatrize wound [3], and for the treatment of leprosy, dysentery, anemia, hemoptysis, debility, stiffness, tonsillitis and skin diseases [2]. The leaves are often crushed and the juice released to treat generalized skin diseases such as scabies, ringworm, rash and wound, while the stem or stem twigs are employed as chewing sticks for oral hygiene [4]. When chewed with kola nut, its leaves serve as an antiemetic, while its leaf sap could be used for treating ear infections, fever, headache, malaria and cephalgy [5]. The leaves are made into plaster to treat snakebite in Southwest Cameroon [6]. The seeds are used traditionally as a remedy for stomach problems like diarrhoea, dysentery, jaundice [7].

Kidney disorders are one of the world problems. Despite its frequent occurrence, high morbidity and high mortality, its medical management is currently inadequate. The association of fulminant renal failure with diseases of the liver and the biliary tract is known for more than a century and has already been reported, in 1863 by Austin Flint, in a case series of patients with cirrhosis and ascites [8]. However, no effective treatment that delays disease progression and complications has yet been found. Several recent studies suggest that traditional herbs and micronutrients such as carotenoids and selenium may be useful for this purpose [9].

The seeds of *DE* are used traditionally as a remedy for gastrointestinal disorders like diarrhoea, dysentery [7]. The aim of this study is to ascertain the effect of the ethanolic extract of *DE* on some biochemical kidney function (creatinine and urea concentration), electrolytes (sodium, potassium, chloride and bicarbonate) and histological structure of the kidney in carbon tetrachloride (CCl_4) induced renal toxicity using adult male Wistar rats as the animal model.

2. MATERIALS AND METHODS

2.1. Plant material and preparation: The *Dacryodes edulis* seeds were collected around Awka, Anambra State, Nigeria during May - June and authenticated by a Botanist, in the Department of Botany, Ambrose Alli University, Ekpoma, Edo State, Nigeria. The seed were shade-dried and thereafter pulverized. 850g of the pulverized seeds was soaked in 70% ethanol for 4 days. The ethanol mixture was filtered to obtain the ethanolic extract according to the method of Akinpelu [10]. The residue was re-extracted, the same solvent in 24 hours. The mixture was concentrated in-vacuo using a rotary evaporator to remove the ethanol [11]. The resultant was then used for the experimental work.

2.2. Study design: Adult male Wistar rats (weighing 180-200g) were sourced from the animal house, College of Medicine, Ambrose Alli University, Ekpoma. They were housed in clean cages and fed with normal pellet and water *ad libitum*. They were acclimatized to the laboratory conditions for 2 weeks. The rats were randomly divided into 3 groups as normal control, negative control treated with 1ml/kg body of CCl_4 in olive oil (at a ratio of 1:1; intraperitoneally injection) every 72 hours for 3 doses and test group receiving 1500 mg/kg *Dacryodes edulis* extract (via intragastric route) after CCl_4 intoxication. Six rats were sacrificed from each group on days 1, 7, 14, 21 and 28 post treatment.

2.3. Estimation of biochemical parameters: Sodium, potassium, chloride and bicarbonate ions where estimated using the ion selective electrode machine. Serum urea was estimated using photometrically by Berthelot's reaction. Serum creatinine was estimated via colorimetrical method using Randox kits.

2.4. Histopathological studies: The kidney was harvested and fixed in 10% neutral buffered formalin for histological processing. The slides were viewed at an appropriate objective to check for the architecture of the tissues. The pictures were displayed in photomicrographs and the histology was analyzed by a histologist with the University of Medical Sciences, Ondo State.

2.5. Statistical analysis: The results were expressed as Mean ± SEM. Data were analyzed using Statistical Package for Social Science (SPSS/Version 17.0) software. Significance between experimental groups was determined using one-way analysis of variance (ANOVA) followed by Least Significant Difference (LSD) test for comparison between two groups. The p-values less than 0.05 were considered statistically significant.

3. RESULTS

1. The effect of the ethanolic seed extract of *DE* on serum kidney functions in CCl_4 induced renal damage in adult male rats

CCl_4 exposure resulted in a significant increase in serum urea and creatinine levels at baseline. Treatment with *DE* resulted in significant decrease in the CCl_4 induced increase in serum urea level by 7 days and serum creatinine level by 14 days (Figure 1).

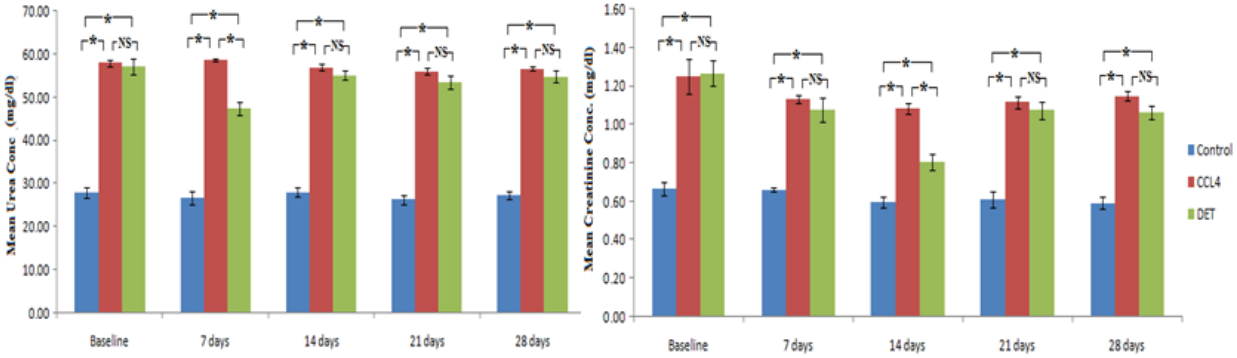


Figure 1. Effect of ethanolic seed extract of *DE* on serum urea and creatinine concentration in CCl_4 induced renal damage in adult male rats.

Values are mean ± Standard error of mean (SEM), *significant difference, NS = Not significant, CCl_4 = untreated carbon tetrachloride exposed, DET= *Dacryodes edulis* treated.

2. Effect of ethanolic seed extract of *DE* administration on serum electrolytes in CCl_4 induced renal damage in adult male rats

Figure 2 shows the effect of ethanolic seed extract of *DE* administration on serum electrolytes in carbon tetrachloride induced renal damage in adult male rats. CCl_4 exposure caused a significant increase ($p<0.05$) in serum Na^+, K^+ and Cl^- concentrations but no significant effect on serum HCO_3^- concentration at baseline. Compared to the control, serum Na^+ were significantly higher ($p<0.05$) in the untreated exposed and *DE* treated groups throughout the duration of treatment. Serum K^+ and Cl^- were not significantly decrease in *DE* treated group compared to the untreated exposed group and was significantly higher compared to the control. similarly, treatment with *DE* did not result in any significant change in serum HCO_3^- compared to the control or the untreated groups.

3. The effect of the ethanolic seed extract of *DE* on the histology of the kidney in CCl_4 induced renal damage in adult male rats.

Slide A shows normal cytoarchitectural features with distinct glomeruli (white arrow). Slide B presented vacuolations (hexagonal enclosure) and defective glomeruli (blank arrows). Slide C presented defective/shrunken glomeruli (stared arrows) as compared to the relatively normal glomeruli (block white arrows).

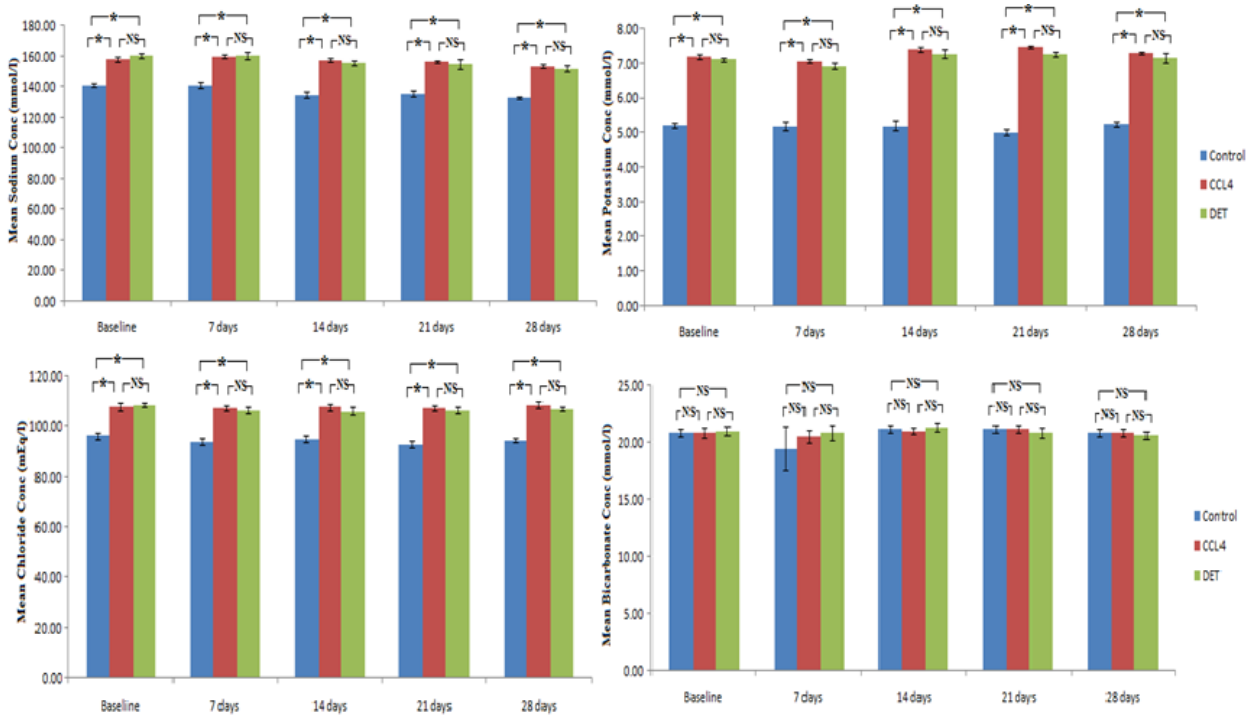
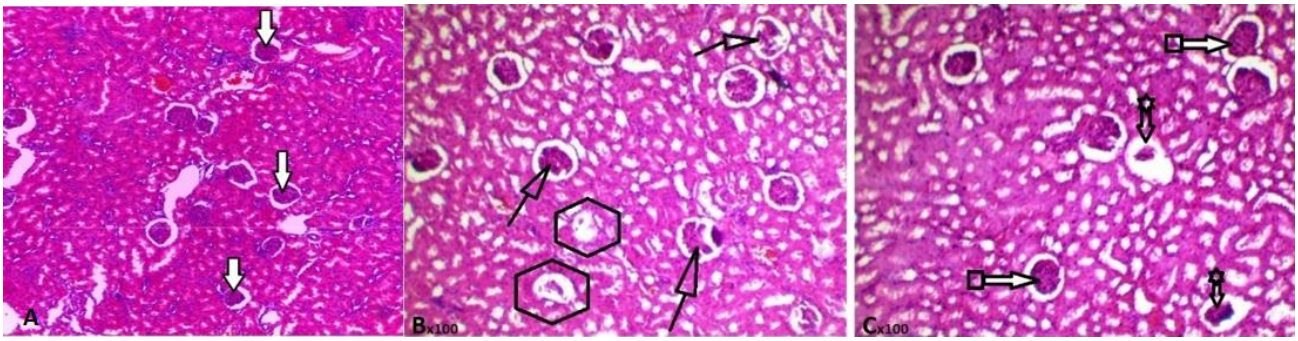


Figure 2. Effect of ethanolic seed extract of *DE* on serum sodium, potassium, chloride and bicarbonate concentration in CCl_4 induced renal damage in adult male rats.
Values are mean ± Standard error of mean (SEM), *significant difference, NS = Not significant, CCl_4 = untreated carbon tetrachloride exposed, DET= *Dacryodes edulis* treated.



Plates 3: Kidney tissue sections –A (control), B (CCl_4) and C (DET) (H&E x100).

4. DISCUSSION

The Liver is not the only target organ of CCl_4 as it also affects several organs of the body including the kidneys [12] through oxidative stress cellular/tissue/organ injury and carcinogenesis [13]. Considering that CCl_4 triggers protein oxidation and lipid peroxidation to produce these damages (14-16], the effect on the kidney was investigated

in this study. As a measure of renal function status, serum urea and creatinine are often regarded as reliable blood markers [17]. Elevations in the serum concentrations of these markers are indicative of renal injury [17-19]. This is because the kidney is responsible for filtration of the blood, and when there is an injury to the kidney, urea and creatinine which are supposed to be filtered into the urine accumulates in the blood [18, 19]. In the present study, the concentration of urea and creatinine in the serum of rats following CCl_4 exposure increased. The observed increase in serum urea and creatinine indicate renal injury which was supported by histological changes. Olagunju *et al.* [20] has observed CCl_4 to induce a significant elevation in the levels of serum uric acid, urea and creatinine levels within 72 hours of exposure. In the present study, treatment with DE resulted in significant decrease in the CCl_4 induced increase in serum urea in 7 days but there was no significant difference by 14, 21 and 28 day of treatment. On the other hand, DE treatment resulted in significant decrease in the CCl_4 induced increase in serum creatinine only at 14 days treatment but the level was still significantly different from the control.

Treatment of CCl_4-induced kidney damage with ethanolic seed extract of *DE* did not follow a clear pattern; however, the comparable serum urea and creatinine levels following treatment with ethanolic seed extract of *DE* indicate that the seed has a poor effect in resolving the damage to the kidneys and thus a poor reno-protective agent. In consonance with this observation, Okolo *et al.* [21] has documented the hexane extract of *DE* at administered dose of 1600 mg/kg, positively regulated glycaemia, restored the desirable lipids but did not adversely affect kidneys function as observed by the creatinine and urea results.

Although the literature reported medicinal plants with nephroprotective properties to mediate their protection via antioxidant activities and high concentration of flavonoids and alkaloids content [22, 23]. Although saponins have been reported to protect the kidneys against carbon tetrachloride intoxication [24], the protective effect on the kidney was not observed in this study probably due to its low concentration in the seed. Summing the observed effect of urea and creatinine in the study, it is plausible for the alkaloid and flavonoid components of ethanolic seed extract of *DE* not to be effective in restoring kidney injury by CCl_4.

On serum electrolytes, the present study showed that CCl_4 exposure caused an alteration from the baseline level of the serum electrolytes with the potentials to increase serum $Na+$, K^- and Cl^- and HCO_3^- concentrations. The alterations of these electrolytes by CCl_4 exposure indicate change in kidney function and as such suggest renal injury. These findings corroborate previous data on CCl_4 induction of renal damage [25]. The damage of CCl_4 to renal tissue may call for critical strategies to protect the kidneys of people who are occupationally or otherwise exposed to CCl_4. In the present study, treatments with DE did not significantly diminution serum induced electrolytes increased by CCl_4. It is known that sodium, potassium and chloride ions regulate the fluid balance in the body [26]. Thus, the observed CCl_4 induced alterations is also likely to disturb fluid balance.

Conclusion

The findings of this study did not show a strong possible restorative effect of *DE* on CCl_4 induced renal damage. However, there is need for further research using higher doses and probably in-vitro studies. Hence, the reno-protective potential of *DE* is in doubt.

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